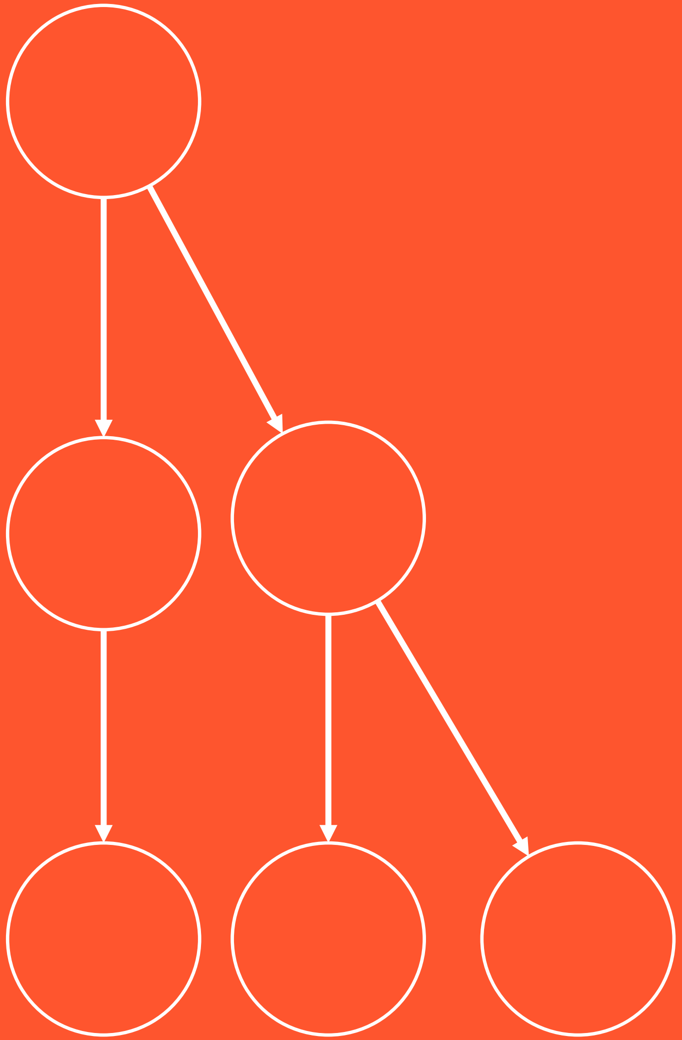


BCOG 100

Social Cognition
Lecture 11.2 — Predicting Others and Learning from Them

UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

To Start: What Did You Actually See?

In your group, three minutes.

Think of a time that you correctly predicted what a friend would do next. What did you actually see or hear? What did you have to infer?

Could you have made the same prediction just from what that friend usually does?

A Strategy That Requires Very Little

Tit-for-tat: cooperate on the first move, and then do whatever the partner did on the previous move.

Tit-for-tat: cooperate first, then copy the last move

C = cooperate *D = defect* *moves, in order ->*

Against tit-for-tat

tit-for-tat	C	C	C	C	C	C
tit-for-tat	C	C	C	C	C	C

cooperates on every move

Against always defect

tit-for-tat	C	D	D	D	D	D
always defect	D	D	D	D	D	D

loses only on the first move

Three properties

It never defects first.

It defects in return after a partner defects.

It goes back to cooperating as soon as the partner does, after only one defection of its own.

What a player needs

Tell partners apart (or stay with one, or encounter them only at one fixed place). Respond to one move, the partner's last.

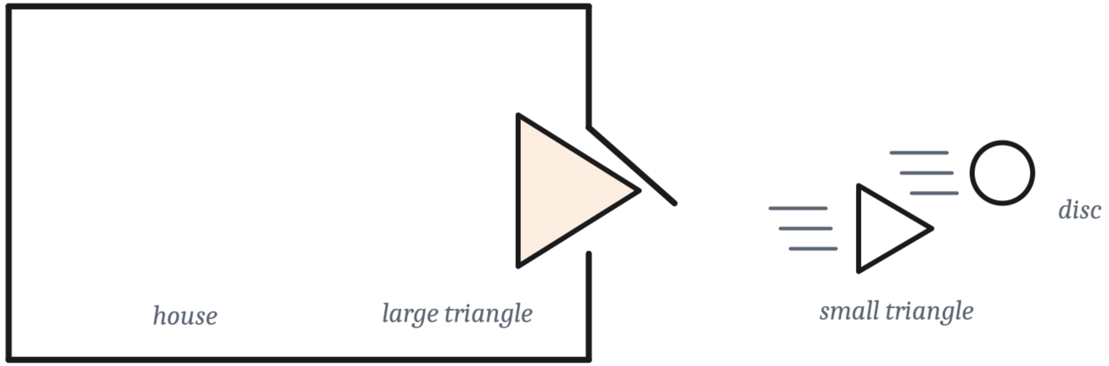
The game must be likely to continue.

No prediction about what a partner thinks or wants: only a response to what it did.

Goals, Gaze, and Moving Shapes

Theory of mind: the ability to attribute states such as goals, perceptions, and beliefs to others, and to predict behavior from them. **Joint attention**: sharing attention to an object with another individual, by following its gaze to what it is looking at and looking back at it.

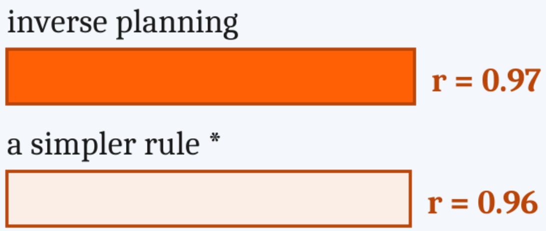
One moment from a film of moving shapes



a schematic redrawing; after Heider and Simmel (1944)

Goal from motion

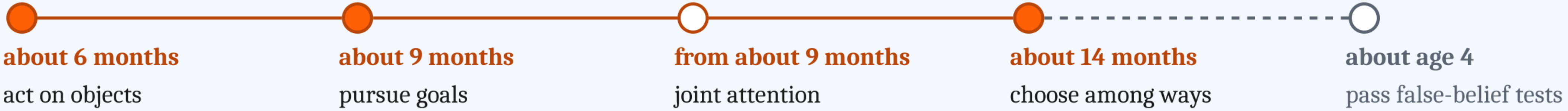
Inverse planning: which goal makes the seen path efficient?



* what it last moved toward

Differed only on ambiguous last steps.

Goals before beliefs, in infants (one synthesis)

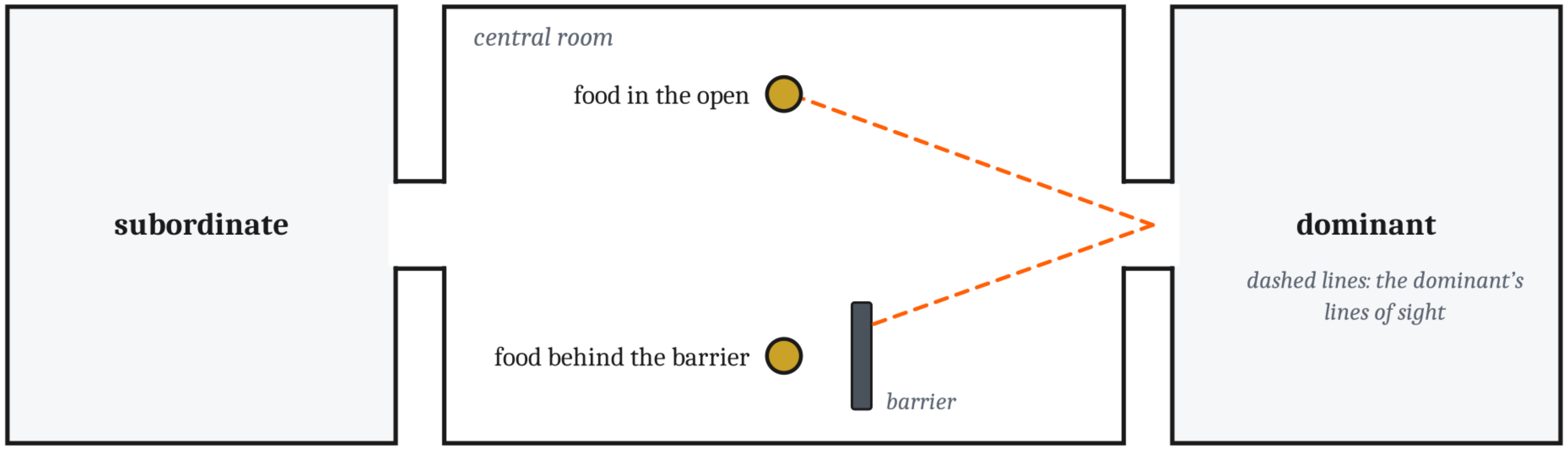


- **about 6 months** act on objects
- **about 9 months** pursue goals
- **from about 9 months** joint attention
- **about 14 months** choose among ways
- **about age 4** pass false-belief tests

What Others Can See

Morgan’s Canon: a behavior should not be explained by a higher mental process if a simpler one accounts for it. So simpler explanations are ruled out first.

Who can see which piece of food



central room
food in the open
subordinate
dominant
dashed lines: the dominant’s lines of sight
food behind the barrier
barrier

top view, not to scale; after Hare et al. (2000), Fig. 5

A 2008 review

Strong evidence: chimpanzees understand others' goals, and what others see and know.

No evidence: that they understand a false belief.

Left open:
seeing, or behavior?

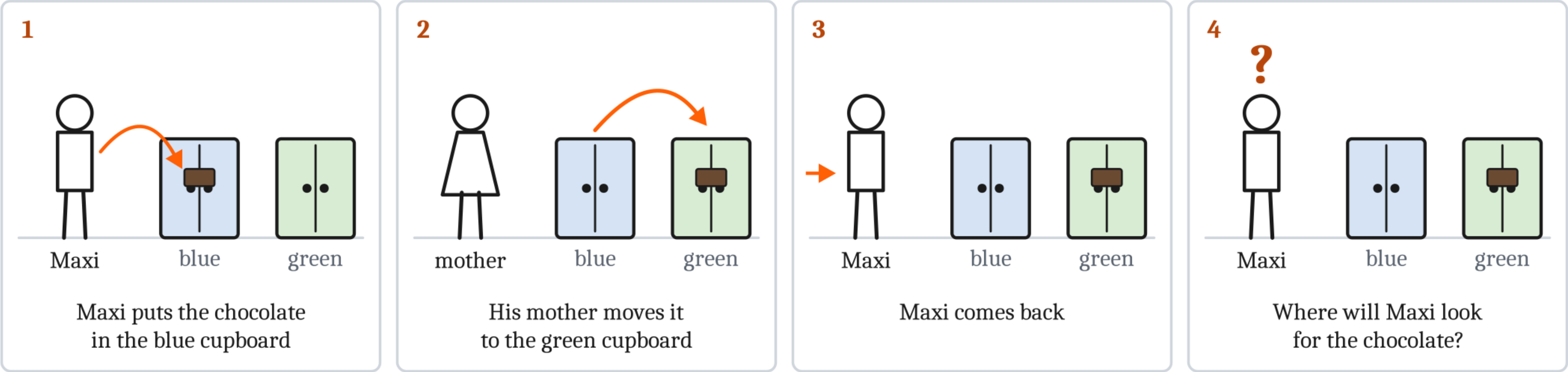
Simpler explanations ruled out, in turn

1. ~~food only when no dominant is present~~
2. ~~only what the dominant cannot reach~~
3. ~~following the dominant's movements~~
4. ~~avoiding where the dominant looked~~
5. ~~a preference for food near barriers~~
6. ~~only what the dominant sees now~~

False Belief, With Words and Without

False belief: a belief that does not match the way the world is, such as a belief that an object is where it was left after it was moved.

The Maxi story


1
Maxi
blue
green
Maxi puts the chocolate in the blue cupboard
2
mother
blue
green
His mother moves it to the green cupboard
3
Maxi
blue
green
Maxi comes back
4
?
Maxi
blue
green
Where will Maxi look for the chocolate?

after Wimmer and Perner (1983), Table 1

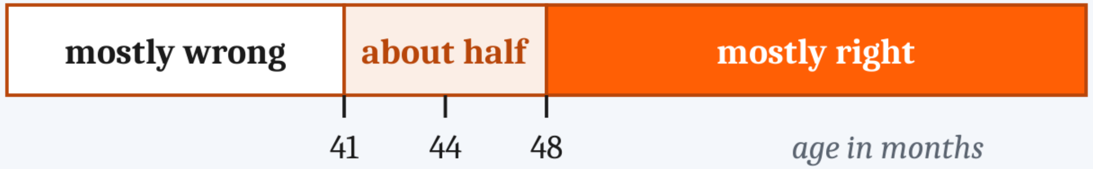
Language

A strong correlate of passing.

But every task was itself verbal.

When children pass

178 studies, more than 4,000 children



Without words: infants and apes

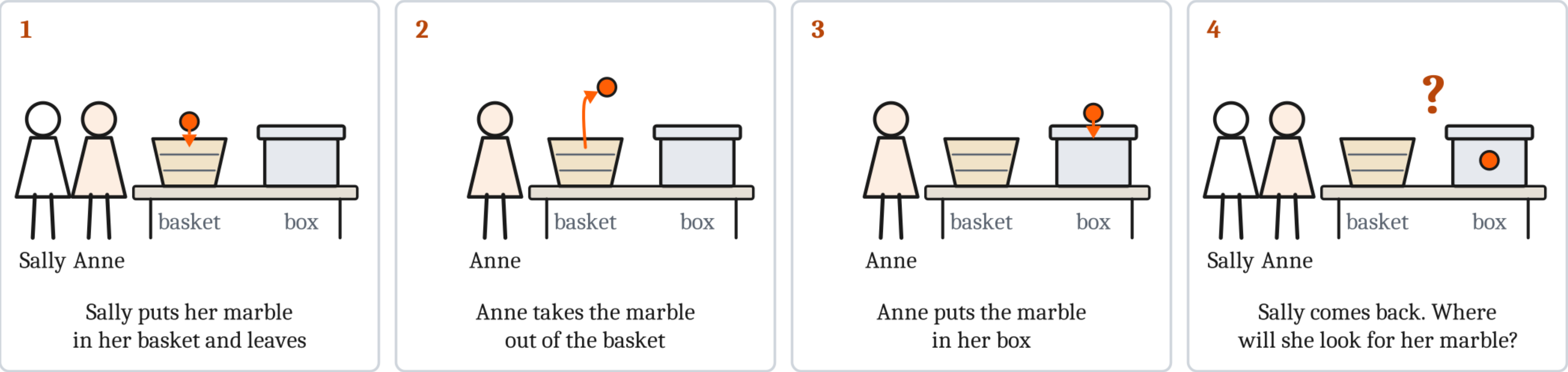
Longer looking, or looking first, where a fooled person would search.
Hard to repeat; results mixed.

An open question.

Autism, and Misunderstanding in Both Directions

Double empathy problem: the proposal that difficulties in understanding between autistic and non-autistic people are mutual, rather than located in one person.

The Sally–Anne task



Sally puts her marble in her basket and leaves

Anne takes the marble out of the basket

Anne puts the marble in her box

Sally comes back. Where will she look for her marble?

after Baron-Cohen, Leslie, and Frith (1985), Fig. 1

Read narrowly

Most of 20 autistic children pointed to the box: one task, groups that differed in language.

Later: many autistic people pass.

A story retold along chains of eight people

autistic	A ▸ A ▸ A ▸ A ▸ A ▸ A ▸ A ▸ A	as many details passed on
non-autistic	N ▸ N ▸ N ▸ N ▸ N ▸ N ▸ N ▸ N	as many details passed on
alternating	A ▸ N ▸ A ▸ N ▸ A ▸ N ▸ A ▸ N	**fewer; most lost at the first handoff**

A = autistic
N = non-autistic

a small study; needs to be repeated

Five Accounts of Predicting Others

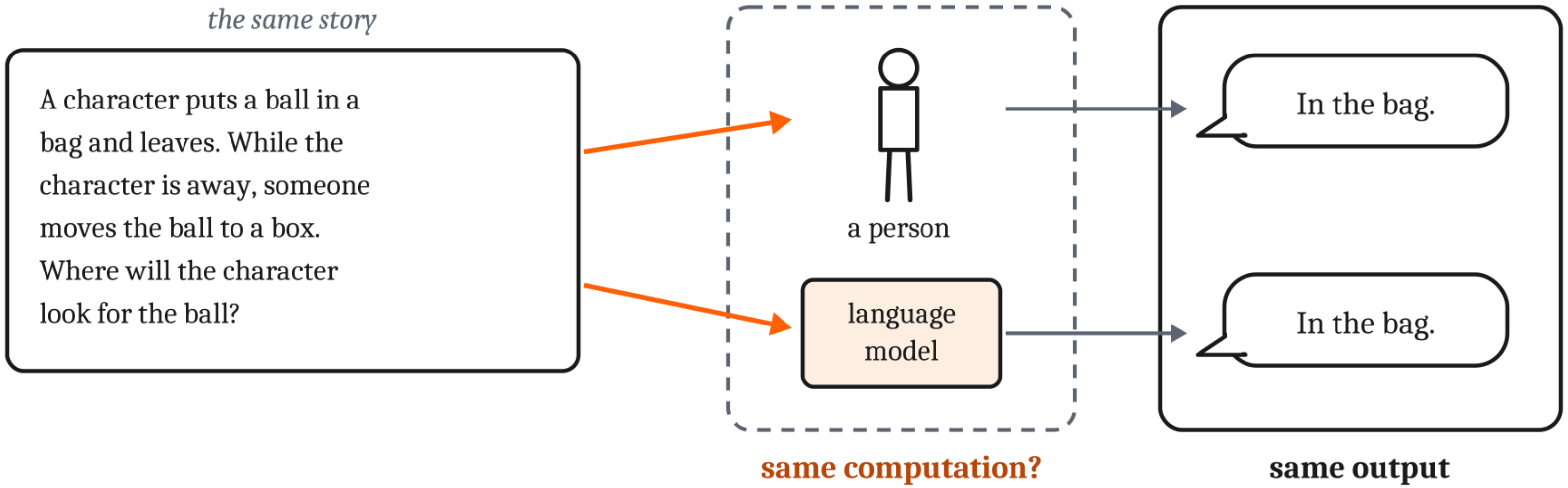
The five accounts differ in what each says the predictor represents. Simplicity does not decide among them; specific tests might.

Account	What the predictor represents	A test that could tell it apart
Theory	Others’ unobserved mental states, and rules for reasoning from them	Correct predictions in new situations, without a long history of observing similar ones
Simulation	No theory; its own decision-making, run offline with the other’s situation as input	Whether its predictions of others match how it would itself act in the same situation
Behavior-reading	Regularities in behavior, and no mental states	Success only with behaviors seen often, and no confusion between what it sees and what another sees
Submentalizing	Nothing about minds; attention and other general processes produce the effect	Inanimate controls, such as an arrow in place of an avatar, produce the same effect
Two systems	A fast tracker of what another last encountered, beside slow reasoning about beliefs	Fast success on where an object is, but failure on more complex contents of a belief

Machines on Theory-of-Mind Tests

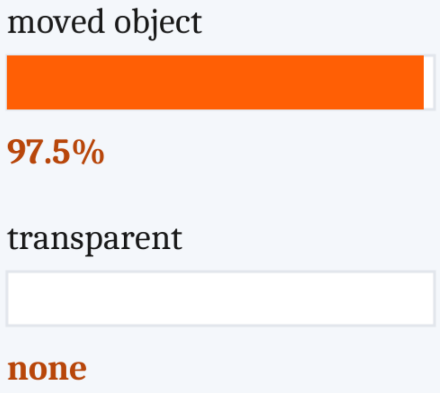
Since about 2023, the largest language models have answered standard written false-belief questions about as well as adults do. Passing a test is not the same as representing others’ mental states.

The same answer, but the same computation?


the same story
A character puts a ball in a bag and leaves. While the character is away, someone moves the ball to a box. Where will the character look for the ball?
a person
language model
In the bag.
In the bag.
same computation?
same output

after Hu, Sosa, and Ullman (2025), Fig. 1

GPT-4, one test



What such failures mean is disputed.

Two separate questions

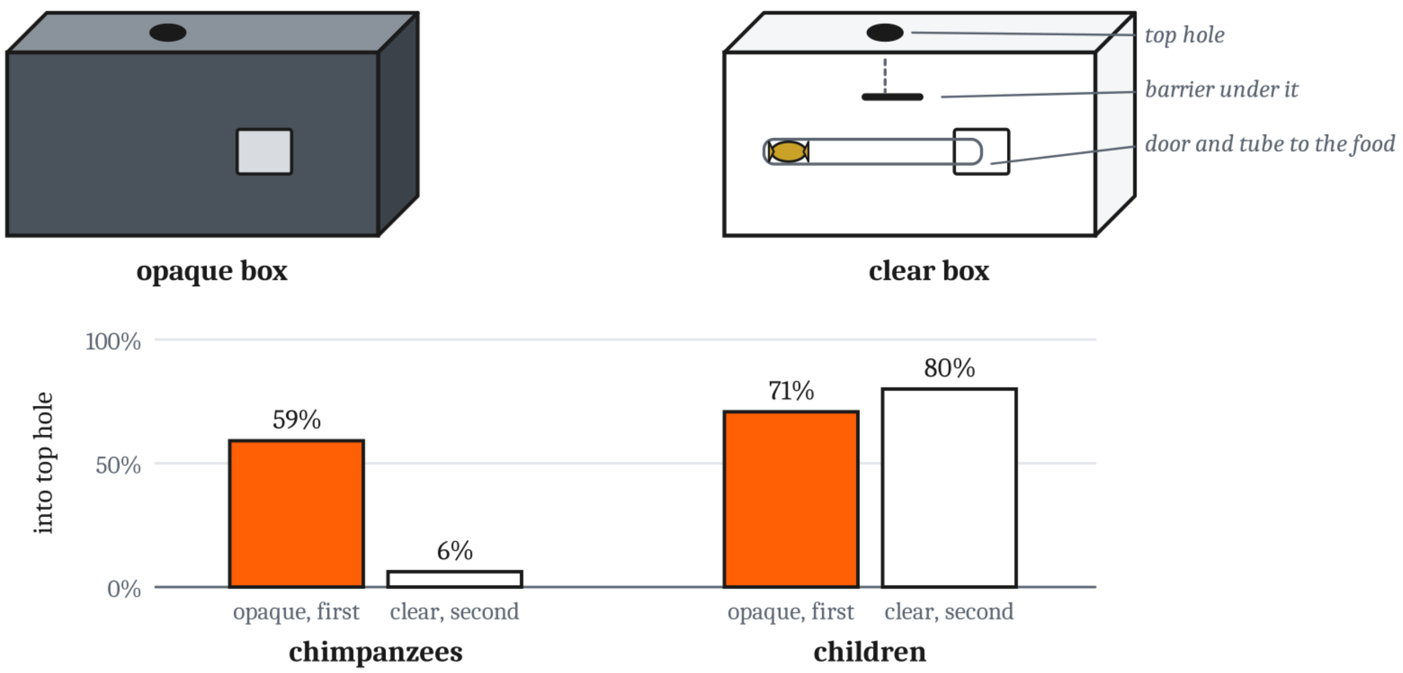
Passing a test: giving the answers that someone representing others' mental states would give. It does not mean reaching them that way.

Predicting a partner well is also not the same as responding well to one.

Learning from Others

Social learning: learning that is influenced by watching, or interacting with, another animal. **Overimitation**: copying actions that are plainly unnecessary for reaching a goal.

Copying actions that do nothing



median share of stick insertions into the top hole, for the groups given the opaque box first; after Horner and Whiten (2005), pp. 169–176

What is reproduced

stimulus enhancement
drawn to a place

observational conditioning
the other's response

emulation
the result

imitation
the form of the actions

less to more of the behavior

Teaching: changing behavior near a naive animal, at a cost or for no gain, so that it learns faster (meerkat helpers)

Evolutionary Psychology: The Program

Evolutionary psychology: a research program for testing proposals that features of the human mind are mechanisms, each specialized for a particular problem that human ancestors faced.

The founders’ principle	Our stance in this course
The brain functions like a computer	Close to the perspective used in this course, and to Marr’s levels
Its circuits were shaped by natural selection for ancestral problems	Few deny that evolution shaped the brain; each specific circuit needs testing
Most of what goes on in the mind is hidden from consciousness	Uncontroversial in this course
Different circuits are specialized for different problems	Consistent with the course, but to be applied carefully; modularity is a matter of degree
“Our modern skulls house a stone age mind”	The most tenuous: a mind that learns from its present environment is less likely to be stranded in an ancestral one

Tested One at a Time, and How Bateman Was Read

Some proposals held up when tested, and others had weaker support than first reported. Scientists’ own beliefs can shape how evidence is read.

Tested one at a time

Kin detection

predicted before the data;
supported, moderately

Faces near ovulation

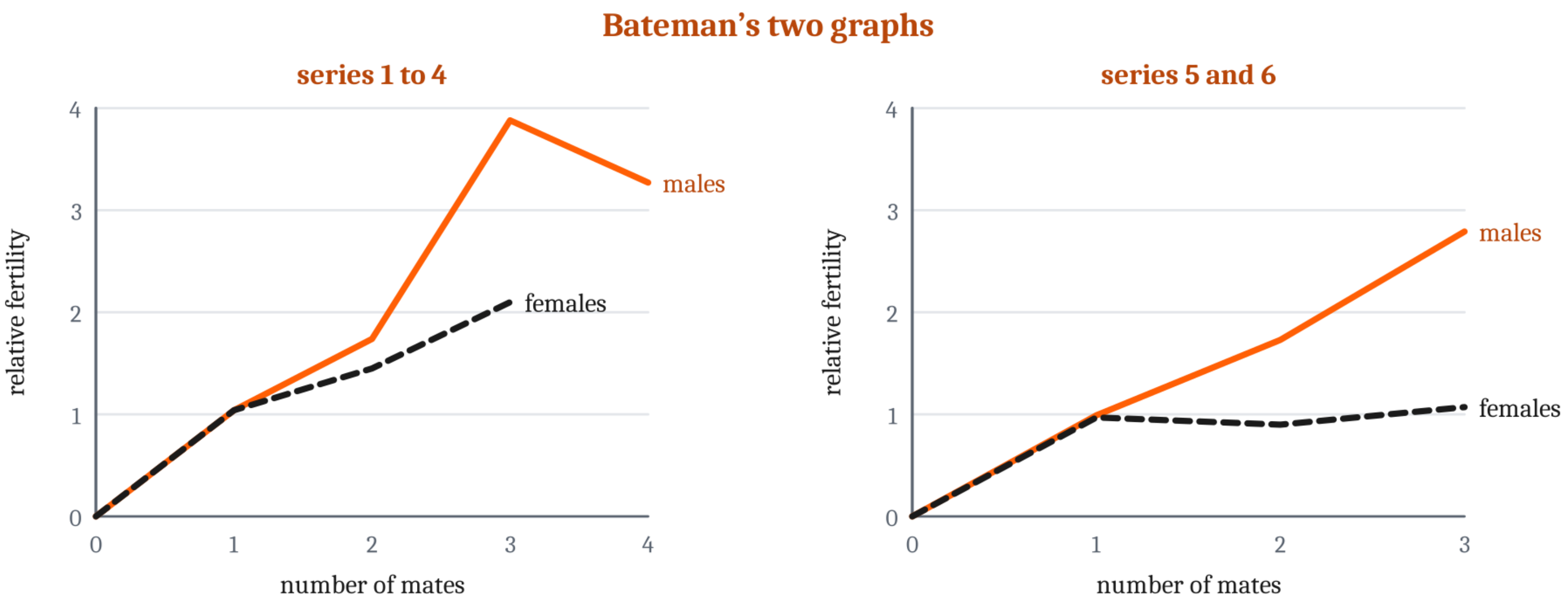
a large study found
no support

Partner preferences

stated ideals differ by sex;
face to face, about equal

Initial skepticism

patterns are tendencies,
not rules



read by eye from Tang-Martínez (2016), Fig. 1, which reproduces Bateman (1948); approximate

How Bateman was read

His method was biased, so it cannot test his claim. In the graph from most of his data, both sexes gained with more mates; textbooks mostly showed the other.

Stereotypes as Categories

Stereotype: a set of beliefs about the typical members of a social group, applied to an individual because of the group the individual belongs to.

Warmth

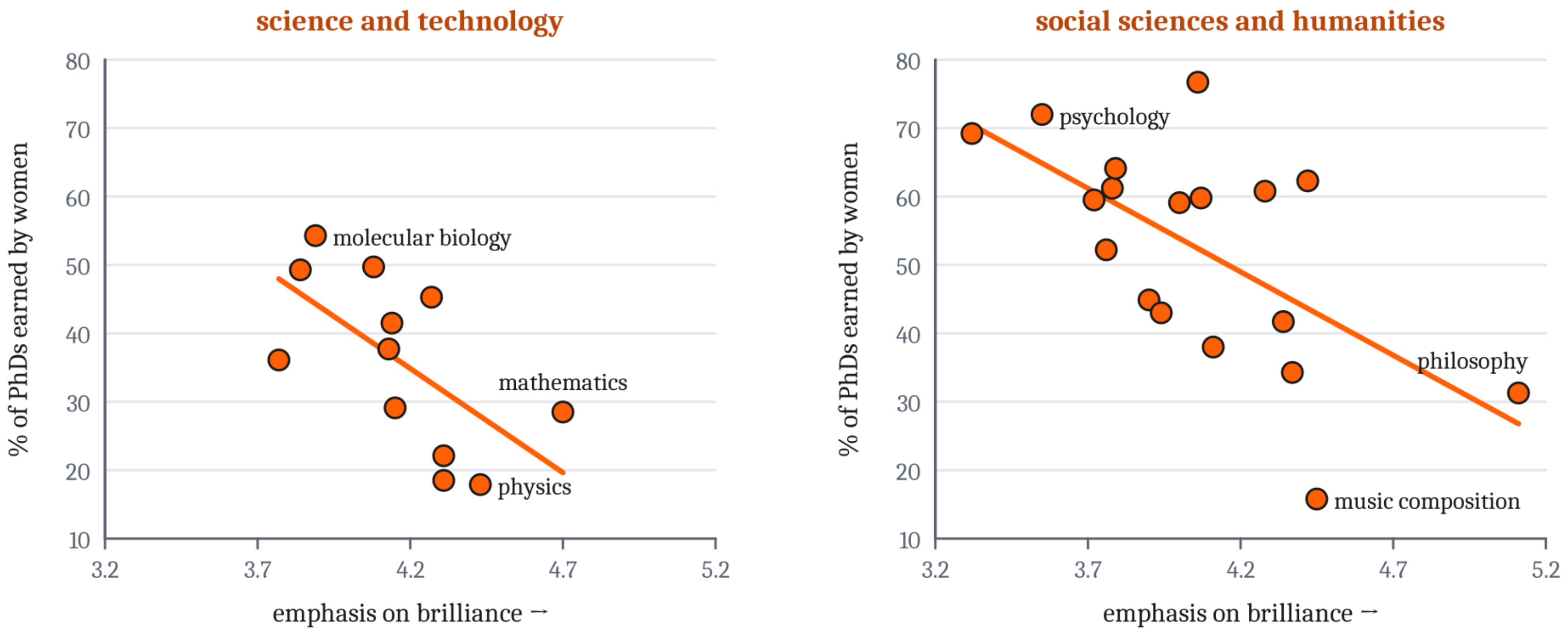
	competence *low*	*high*
warm	**Pity**	**Admiration**
not warm	**Contempt**	**Envy**

the emotion each tends to draw

Gender

78% of measured differences: small or close to zero.
By 6 and 7, girls no longer favored their own gender; boys still did.

Beliefs about brilliance and women's share of PhDs


science and technology | **social sciences and humanities**

each dot is one field; after Leslie, Cimpian, Meyer, and Freeland (2015), Fig. 1

r = -0.60, across fields: brilliance beliefs may shape fields; rivals compared, not ruled out.

To Discuss: Does the Chatbot Have a Theory of Mind?

In your group, five minutes.

A headline says that a chatbot “has a theory of mind” because it passed the Sally–Anne test.

What would we want to know before agreeing? Write an altered version of the story that a person tracking Sally’s belief would still answer correctly.

What would a researcher applying Morgan’s Canon to apes ask about the chatbot?

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